# Syllabus

**STAT 141: Introduction to Probability and Statistics**
**Reed College**
**Fall 2026**

Professor Megan Ayers

## Useful Information

**Lecture and Lab Instructor**: Megan Ayers (she/her)

- Email: meganayers@reed.edu
- Office hours: See Moodle for weekly office hour times and locations. I am also happy to meet ad hoc by appointment if you have conflicts during the scheduled times. Please email me or send me a Slack message to find a time to meet.

**Course Assistants**: We have X course assistants for STAT 141 this semester! A course assistant will be present at each lab section. Further, each course assistant will hold office hours during the week. Times and locations of course assistant office hours can be found on Moodle. Feel free to go to any course assistant's office hours, even if they are not the course assistant for your lab section.

**Links and course resources**:

The course website, link-here, includes information on the course, lecture slides, public course materials, and links to all other course resources:

- Textbooks (links on this Syllabus page)
- RStudio Server, for working on homework and lab assignments
- Slack, for course correspondence,
- Gradescope, for turning in assignments, and
- Moodle, (sparingly) for information about meetings and private course materials

Note that **the entire course website (including lecture slide material) is searchable!!** This may come in handy when completing assignments. Click on the magnifying glass icon in the upper right corner of the website to search.

**Meeting Times**:

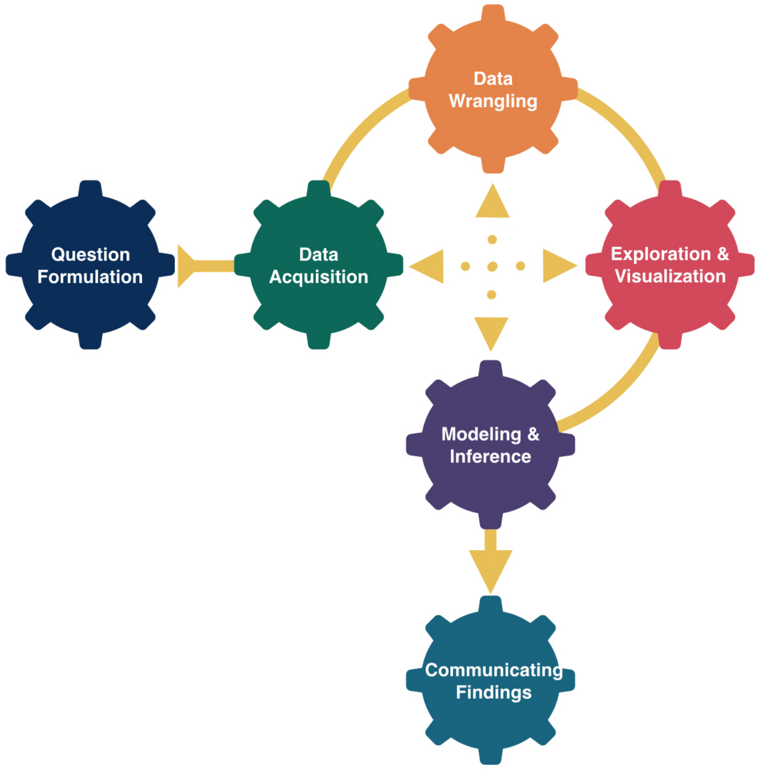
We'll have a lecture-style meeting three times a week, and a lab meeting once a week. Please see the course Moodle page or the Reed schedule of classes to verify the times and locations for your section.

**Communication**: I am most easily reached over email on weekdays between 9am and 5pm. I will do my best to answer your email within one school day. *For urgent matters, please email me rather than sending a Slack message.* For non-urgent course-related questions, particularly questions that fellow classmates or course assistants may know the answer to, your question may be answered more quickly in a public channel on Slack, which I will also monitor.

## Learning Outcomes

In this course, you will learn how to think critically with data by engaging in the entire data analysis process through theory, computation, and practice. While most of our time will be spent learning techniques related to the *Exploration and Visualization* step and the *Modeling and Inference* step, you will also practice the other important components of analyzing data. Furthermore, since computation is an integral part of modern statistical work, you will learn to write `R` code – leveraging the `RStudio` user interface – to analyze data and will become familiar with several tidyverse `R` packages. You will also develop a reproducible and shareable workflow by using `Quarto` documents for all analyses.

By the end of the course, you will have improved your data acumen and your ability to think statistically. More concretely, you will be better able to accomplish the following tasks, which have been broken down by steps of the data analysis workflow:




**Question formulation:**

- Translate a research problem into a set of questions that can be answered with data.
- Formulate data questions as measurable statements about statistics and parameters.

**Data acquisition:**

- Determine the necessary data to conduct analyses.
- Reflect on how design structures and data collection impact potential conclusions.
- Identify potential ethical concerns surrounding data collection and data privacy.

**Data wrangling:**

- Explore datasets to determine what wrangling may be required (e.g., removing missing values, filtering out variables or observations, collapsing categories of a categorical variable).
- Apply basic data wrangling operations.

**Exploration and Visualization:**

- Understand key principles of designing and creating effective data visualizations.

- Create graphs from data, and appropriately draw conclusions from graphs.
- Compute and interpret summary statistics.

**Modeling and Inference:**

- Understand and be able to explain key probabilistic and inferential concepts, such as, sampling, variability, random variables, distributions, confidence, and significance.
- Use probability theory and data to recognize patterns and associations between variables.
- Create and assess the appropriateness of linear regression models for a given problem and set of data.
- Appropriately apply and draw inferences from a statistical analysis using both simulation-based and theory-based methods, including quantifying and interpreting the uncertainty in model estimates.
- Consider the ethical implications of various modeling practices.

**Communicating Findings:**

- Develop a reproducible workflow using Quarto documents.
- Interpret and communicate results of statistical analyses effectively for both a statistical and a non-statistical audience.
- Be able to reflect on the data involved in an analysis and show a curiosity for other ways of examining and thinking about the data.

## Learning Materials & Tools

**Textbooks**: We'll use three textbooks for our course, all of which are freely available online:

- Statistical Inference via Data Science: A ModernDive into R and the tidyverse, Second Edition
  - Reading abbreviation: **MD**
- Introduction to Modern Statistics, Second Edition
  - Reading abbreviation: **IMS**
- OpenIntro Statistics, Fourth Edition
  - Reading abbreviation: **OI**

**Technologies (R, RStudio, and Quarto)**: R is a free and open source programming language, RStudio is an Integrated Development Environment (IDE) which allows for streamlined use of the R programming language, and Quarto is a markdown language that allows for reproducible documents that include R code, text, images, and much more! We will access these technologies via the RStudio Server for this course. A laptop that can access the internet and use the RStudio Server is required for this course.

**Please let me know ASAP if you do not have access to a personal computer!**

## Assignments, activities, and exams

We'll have a variety of assignments, activities, and assessments for this course. In particular:

- **Lab assignments**:
  - Every week we will have a lab assignment. The lab assignments will be worked on collaboratively during lab time on Thursdays, where you will be able to ask the instructor and the course assistant questions. They are designed to help you practice implementing concepts and tasks from lecture, particularly those that relate to coding in R. The following homework assignment will build on questions covered in lab.

  - Lab assignments will be made available for download from the course website before each lab.
  - You will submit the progress you made on each lab assignment via Gradescope at the end of your section's lab time. Labs will be assessed as part of your participation grade, and while you may receive some feedback on your lab work, you will not receive a numeric score.

  - Late labs will not be accepted, though you are expected to complete them on your own to prepare for homework assignments even if you miss lab.
  - Lab assignments will count for a small portion of your grade, but will be very important for preparing for the upcoming homework assignment and quiz.
- **Homework assignments**:
  - Every week we will have a homework assignment. Homework assignments will be posted on the course website on Thursday mornings.
  - These will be turned in via Gradescope by **9:00am on Thursay of the following week**.
  - Homework assignments will largely build on or continue content from that week's lab, and may also cover additional material from lecture.
  - You will receive detailed feedback on your homework assignments, and scores will be assigned based on completion and effort.
  - Homework assignments represent a higher proportion of your grade than labs, but a relatively small proportion overall.
  - Engaging with homework assignments and discussing them in office hours will be the best way to study for the next week's quiz!
  - Completion grades and feedback are given based on your submitted PDF, and regrade requests will not be accepted. Be sure to check that your rendered PDF

legibly displays all of your work before submitting homework assignments to avoid formatting issue frustration!

- **Quizzes**:
  - Almost every week at the beginning of lab we will have a short quiz, based closely on the material from the last week's homework assignment.
  - Cumulatively, quizzes have the largest impact on your grade. However, there will be 10 quizzes throughout the semester, so each individual quiz will be relatively low-stakes.
  - Quizzes will be taken on paper and will be entirely closed-notes.
  - While you will not be asked to write code in R during quizzes, you may be asked to write pseudo-code, describe coding steps qualitatively, or explain code that is shown to you.
  - You will have the opportunity to submit quiz revisions within 1 week of feedback being posted, and to submit make-up quizzes (in the event of absences) for partial credit.
  - Your two lowest quiz grades will be dropped, but dropped quizzes may not include quizzes that you were absent for *if you did not submit a make up quiz.*
- **In-class activities**:
  - Lectures and labs will often involve some in-class activities. Including but not limited to:
    * group activities,
    * independent activities,
    * participation-based learning assessments.
- **Readings**:
  - Readings will be posted ahead of every lecture. These may include a mix of content from the course textbooks and content from past lectures to review.
  - Readings are highly encouraged for deep engagement with course material and commitment of concepts to long-term memory.
  - The content in readings may seem repetitive with lecture content - this is by design! The more frequently you encounter and think through concepts that are presented in different ways, the better you will remember them, and the more deeply you will be able to engage with future material in the course that builds on them.
- **Final Exam**:
  - We'll have one exam (a final exam) for this course. The exam will consist of an in-person written exam.
  - The final exam period will be three hours, and will be held during finals week (the official finals schedule is TBD).
  - No late exams will be accepted.

### Attendance

Being actively present in lectures and labs is very important in this course. This requires attendance, and can look like:

- Active listening
- Contributing to small group or classroom level discussions
- Contributing to discussions on Slack.

Being consistently absent, late to class, or being consistently distracted during class (ex. using cell phone or doing homework during lecture), or consistently not participating in group work will have a negative impact on your participation grade.

### Distribution Requirements

This course can be used towards your **Group III, "Natural, Mathematical, and Psychological Science"** requirement. It accomplishes the following learning goals for the group:

1. Use and evaluate quantitative data or modeling, or use logical/mathematical reasoning to evaluate, test or prove statements.
2. Given a problem or question, formulate a hypothesis or conjecture, and design an experiment, collect data or use mathematical reasoning to test or validate it.
3. Collect, analyze, and interpret data.

This course does not satisfy the "primary data collection and analysis" requirement.

### Course Climate

We expect everyone in this class to strive to foster a learning environment that is equitable, inclusive, and welcoming. If you experience any barriers to learning, please come to Professor Megan Ayers or a college administrator with your concerns.

**Code of Conduct**:

We expect all members of Math 141 to make participation a harassment-free experience for everyone, regardless of age, body size, visible or invisible disability, ethnicity, sex characteristics, gender identity and expression, level of experience, education, socio-economic status, nationality, personal appearance, race, religion, or sexual identity and orientation.

We expect everyone to act and interact in ways that contribute to an open, welcoming, inclusive, and healthy community of learners. You can contribute to a positive learning environment by demonstrating empathy and kindness, being respectful of differing viewpoints and experiences, and giving and gracefully accepting constructive feedback.[1]

[1] This Code of Conduct is adapted from the Contributor Covenant, version 2.0.

## Policies

**Technology policy**:

- For some parts of the course, we will ask you to use your phone/tablet/computer to engage with course material (ex. you'll be asked to take attendance via online survey at the start of lecture, and to use your laptop to work on lab assignments and occasionally during lecture).
- We ask that you do not wear headphones at any point during labs or lectures.
- During other parts of the course, we will ask you to put away all devices, including phones, laptops, and headphones. This will primarily apply to time when the instructor is actively lecturing: please plan to take notes during lectures by hand. Note that all lecture slides will be made available on the course website.
  - **Why this policy?** It is super easy to become distracted by other things on your devices while you are typing notes! App and social media developers are literally trying their very best to distract us all, all of the time. I (Megan) admit that I am easily distracted and am constantly a victim of this. Let's remove the opportunity for distraction during lecture so that we can all focus and engage with the course material.
- Advice for note taking during lecture:
  - You don't need to transcribe everything that is on the slides or that I say, especially because slides will all be posted online. Focus on summarizing content and ideas in your own words, recording any questions you have, and emphasizing any insights that helped you make sense of a concept.
  - Reference the slide number when taking each note so that you can contextualize it with the slides when reviewing your notes. It may be particularly helpful to make list of slides where code examples are provided for different tasks.

**Late work policy**:

- **Quizzes**: Since quizzes are taken in-person during lab, a missed quiz will be assigned a score of 0. However, you have the opportunity to make up a missed quiz outside of class for up to 50% partial credit by submitting a quiz revision. To help with various circumstances expected and unexpected, quizzes that have been made up outside of class are eligible to be dropped from your final grade, no questions asked (the two lowest quiz scores *from attempted quizzes* will be dropped at the end of the semester).
- **Homeworks**: To be considered for full credit, homework assignments must be submitted by their deadline. Late homework assignments will be accepted for up to 50% partial credit at any time before the end of the last day of class, Wednesday December 9th at 11:59pm. Late submissions of homework assignments are not guaranteed timely or

detailed feedback - you are encouraged to meet with Megan or course assistants in office hours to clarify any questions or discuss solutions from any late homework assignments.

- **Labs**: Lab assignments will be due at the end of each lab section and will count towards a small participation grade. You will have a grace period until the end of the day to submit your lab in the case of a rendering error. In the event of a lab absence, you are expected to prepare for homework assignments by completing labs on your own, but late submissions of lab assignments will not be accepted.
- **Exam**: The final exam cannot be made up, except in cases with academic accommodations.

**Collaboration Policy and Academic Honesty**:

Working with your classmates on difficult and interesting problems can not only help your learning, but also help you get to know each other! Therefore, I *highly* encourage you to collaborate on assignments. But, every piece of work you do *must* be your own. Copying other people's work or code is not acceptable (whether they are in the course or not), nor is allowing other people to copy and paste your own work. The Honor Principle must guide your conduct in this class.

If you choose to collaborate with a classmate, **you must add their name to the top of your assignment**, and list them as a collaborator, e.g.:

Collaborator(s): Michael Pearce, Lenny Wainstein, Grayson White

*But what is collaboration?*:

For STAT 141, collaboration can look like: working with classmates together on a given problem, doing scratch work, helping each other get un-stuck on a part of a problem, and even coming to a solution. However, you **must** write up your own problem solutions individually and cannot copy other's solutions (even those who you have collaborated with). You are expected to be able to explain your submitted solutions. Further, copying code from a collaborator, classmate, an internet resource, or a generative AI tool (see the following section) is strictly prohibited.

**AI and Internet Use Policy**:

Artificial intelligence (AI) tools, such as ChatGPT, Claude, Co-Pilot, Gemini, and others are being used to generate code, analyze data, and much more. However, learning to think critically about a problem at hand, and engaging with your peers, tutors, and instructors when not understanding a concept or question are integral components of a liberal arts education. Further, a key goal of this course is for you to learn how to thoughtfully, ethically, and independently extract knowledge from data and engage in statistical reasoning. Therefore, the use of generative AI tools, such as ChatGPT and others, are strictly prohibited in any stage of the work process for this course. If you have questions about whether a tool is allowed for this course, ask the Instructor before using it.

While there are many ways to achieve tasks in R, and many helpful internet resources for learning about them, this course is designed to be self contained, and you may only use functions and packages that have been covered in this course on your assignments. You may use online documentation resources to refresh yourself or learn more about functions that have been introduced in class. If you have questions at any point about other functions or packages, I (Megan) am happy to discuss! If you are unsure of how to approach a particular exercise or of what function to use, help is available to you! Please visit office hours, collaborate with your peers, or post in the Slack channel.

## Acknowledgements

Thank you to Grayson White for allowing me to largely reuse the website structure he developed for Stat 141, and many of the related course materials (which in turn were influenced by the work of many others!). Thank you to Michael Pearce and Lenny Wainstein, many of whose previous Stat 141 course materials have also been adapted for this course. I enlisted the help of Claude Code for lecture slide and assignment reformatting tasks (ex. "Please change Math 141 Spring 2026 to Stat 141 Fall 2026 in all files!", "Please align assignment deadlines in .qmd files with the website schedule", "Please add this collaboration statement to the top of every assignment file").